

Concrete Mix Design Report – M40

Project: High-Rise Structural Works **Concrete Grade:** M40

Material / Parameter	Value	Unit	Remarks
Cement	420	kg/m³	Primary cementitious material
Fly Ash	70	kg/m³	Supplementary cementitious material
Water	160	L/m³	Batch water before moisture correction
Fine Aggregate / Sand	680	kg/m³	Controlled grading required
10 mm Coarse Aggregate	400	kg/m³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	650	kg/m³	Nominal coarse aggregate fraction
Chemical Admixture	4.8	kg/m³	High-range water reducer
Water/Cementitious Ratio	0.327	-	Illustrative value
Target Slump	120	mm	At point of discharge
Target Mean Strength	48.25	MPa	Illustrative 28-day target

Design Notes

The M40 mix is intended as a high-strength illustrative design. Production requires tighter control of aggregate grading, moisture, admixture compatibility, batching accuracy, and strength development.

Document status: Synthetic sample created for RMC RAG development and testing. Not an approved construction mix design. Actual proportions require laboratory validation, material test results, project specifications, moisture corrections, and applicable standards.